Agenthon 2026

T2 Quant V2

EWMA + Student-t Copula Dependence Experiment Report

Core question	V1 improved the marginal and tail scores, but the joint variogram deteriorated. Can we improve the joint score by separating volatility from correlation and replacing historical row bootstrap dependence with Student-t dependence?

What this report should make clear

- What the 24-fold V1 ablation actually improved, and what remained weak
- Why joint dependence became the bottleneck in the final composite score
- How V2 combined EWMA volatility with a Student-t copula-style dependence engine
- What the 2-fold smoke test and the 24-fold validation actually showed
- Why V2 did not solve the problem and why the project moved to a PCA factor model

Version naming note

The initial planning guide used V2/V3 as broad roadmap labels. In the actual implementation process, version names were reassigned to concrete experiments. In this report, V2 means the EWMA + stable Student-t dependence experiment.

1. Starting point - what V1 left unresolved

V1 combined EWMA volatility, a recent/full-history mixture, historical bootstrap shocks, and a weak drift/mean-reversion adjustment. The key 24-fold result was very specific: the individual predictive distributions and tails improved, but the joint cross-tenor structure deteriorated.

Model	Marginal ↓	Joint ↓	Tail ↓	Composite ↓	Win rate vs Ref
Reference Gaussian	1.1858	4.4381	0.1579	1.9559	-
Quant block1	1.1477	4.6657	0.1268	1.9989	41.7%
Quant v1	1.1568	4.8047	0.1358	2.0470	29.2%
Quant no EWMA	1.2046	5.2636	0.1736	2.2161	33.3%

Key interpretation

Quant block1 beat the Reference on marginal score by about 3.2% and on tail score by about 19.7%, but its joint score was about 5.1% worse. Since the joint component carries a 30% weight, that deterioration almost offset the gains elsewhere.

Composite score

$$S = 0.5 \times S_{marginal} + 0.3 \times S_{joint} + 0.2 \times S_{tail}$$

2. What the V1 ablation established

Before designing V2, each major V1 component was removed or modified one at a time. This ablation identified which mechanisms were actually helping.

Experiment	Change	24-fold Composite ↓	Interpretation
quant_block1	5-day block -> 1 day	1.9989	Best Quant result. Five-day serial blocks may have damaged the joint
quant_no_recent	recent_mix 0.60 -> 0	2.0272	The recent pool was not essential, but removing it was not clearly be
quant_no_drift	drift shrink 0.20 -> 0	2.0418	Drift had only a small effect.
quant_v1	Original settings	2.0470	Baseline for the ablation.
quant_no_ewma	EWMA weight 0.70 -> 0	2.2161	Worst result. Regime-sensitive volatility clearly mattered.

V2 design rule Keep the parts of V1 that worked - EWMA volatility and fat-tail behavior - and replace only the dependence mechanism that drives the joint score.

Meaning of EWMA

w_k proportional to lambda^k, $\text{lambda}^{60} = 0.5$ -> 60 days is a half-life, not a hard window

The model still uses the full history; observations simply receive exponentially smaller weights as they become older.

3. Why the joint component was difficult

UST 2Y/5Y/10Y/30Y are not four unrelated assets. They form one yield curve. Even when each tenor has a reasonable marginal forecast, unrealistic relative movements inside a draw can produce a poor variogram score.

One daily shock vector: $\varepsilon_t = (\Delta y_{2Y,t}, \Delta y_{5Y,t}, \Delta y_{10Y,t}, \Delta y_{30Y,t})$

Issue	V1 treatment	Potential weakness
Cross-tenor dependence	Bootstrap an entire historical row	Repeats specific historical curve shapes directly.
Time dependence	Five-day block bootstrap	Short serial patterns may not fit the current regime.
Marginal scale	EWMA + full-history blend	This component was supported by the ablation.

Hypothesis A more stable design may let volatility react quickly while correlation moves more slowly. V2 therefore separated scale from dependence.

4. V2 architecture - EWMA volatility + Student-t dependence

The central change was to split a shock into a scale component and a dependence component.

$$\Delta y_t = D_t z_t$$

D_t is a diagonal matrix of current tenor volatilities, while z_t is a standardized joint shock.

$$D_t = \mathrm{diag}(\sigma_{2Y,t}, \sigma_{5Y,t}, \sigma_{10Y,t}, \sigma_{30Y,t})$$

$$z_t \sim t_\nu(0, R_t), \nu = 7$$

The Student-t distribution preserves heavier tails than a Gaussian. Smaller degrees of freedom correspond to fatter tails.

Correlation stabilization

$$R_{blend} = 0.75\, R_{full} + 0.25\, R_{EWMA}$$

$$R_{final} = 0.95\, R_{blend} + 0.05\, I$$

The final 5% shrinkage toward the identity matrix is a conservative device to reduce sampling noise and extreme estimated correlations.

5. 2-fold smoke test - initially promising

A short 2-fold, 200-draw run was used first to verify that the code and the direction of the experiment were sensible. This was a runtime and sanity check, not a statistical conclusion.

Model	Marginal ↓	Joint ↓	Tail ↓	Composite ↓
Reference Gaussian	1.9434	5.4495	0.4760	2.7017
Quant block1	1.9815	6.1780	0.4394	2.9320
V2 t-copula	1.8997	5.6255	0.4126	2.7200

Initial read

V2 improved marginal, joint, and tail relative to block1, and its composite score was close to the Reference. The joint score moved from 6.1780 to 5.6255, which made the dependence-separation hypothesis look plausible.

However, with only two folds, mean and median coincide and uncertainty is very high. This stage only justified running the full experiment.

6. 24-fold result - V2 did not beat the baseline

The main experiment used 24 historical forecast origins, 21 business-day spacing between origins, and 1,000 Monte Carlo draws per forecast.

Model	Marginal ↓	Joint ↓	Tail ↓	Composite ↓	Win rate vs Ref
Reference Gaussian	1.1858	4.4381	0.1579	1.9559	-
Quant block1	1.1477	4.6657	0.1268	1.9989	41.7%
V2 t-copula	1.1557	4.6812	0.1288	2.0080	37.5%

Decision

The 2-fold test looked encouraging, but the 24-fold result showed that V2 was slightly worse than block1. A simple stable-correlation Student-t dependence engine did not improve the UST curve joint score.

V2 versus the Reference

Component	V2 vs Reference	Interpretation
Marginal	About 2.5% better	Individual tenor distributions remained strong.
Tail	About 18.5% better	The fat-tail calibration advantage remained.
Joint	About 5.5% worse	Still the main bottleneck.
Composite	About 2.7% worse	Did not beat the Reference on average.

7. What V2 taught us - correlation alone was not enough

V2 does not show that copulas are useless. The narrower conclusion is that the joint score in this UST problem appears to depend on more than a single correlation matrix.

Observation	Implication
Removing EWMA caused a large performance loss	Fast-changing volatility regimes matter.
Replacing bootstrap dependence with t-copula did not improve score	A static dependence matrix swap is insufficient.
UST tenors form one curve	A low-dimensional Level/Slope/Curvature factor structure is economically natural.
block1 beat block5	Contemporaneous curve movement may matter more than short serial blocks.

$$\Delta y_t \approx \beta_{1,t} L + \beta_{2,t} S + \beta_{3,t} C + \varepsilon_t$$

L/S/C denote Level-, Slope-, and Curvature-like yield-curve movements. This became the starting point for V3.

8. V2 final checklist

Question	Answer from V2
Which V1 components should be retained?	EWMA volatility and fat-tail behavior.
What was the main V1 weakness?	Joint variogram / cross-tenor dependence.
What changed in V2?	Historical row bootstrap -> stable correlation + multivariate Student-t shocks.
Did V2 succeed?	No. 24-fold composite = 2.0080 versus Reference = 1.9559.
Why did it fail?	UST joint structure appears closer to a curve-factor problem than to a pure correlation proble
What comes next?	Model Level/Slope/Curvature-like factors directly with PCA.

One-line conclusion

V2 preserved the marginal and tail advantages but did not remove the joint bottleneck. The next step is to generate scenarios in yield-curve factor space rather than only modifying covariance or correlation.

Next step

Quant V3: PCA / Yield-Curve Factor Bootstrap. Compare 2-factor and 3-factor structures and test how much residual noise should be restored.